

HHFA indicator tables

All the indicators in a single HHFA table have the same denominator. The HHFA contains different types of tables that can be described according to the indicator inventory dimension (see below), the presence of indices, and the denominator used.

- **General service availability tables** do not contain indices. The denominator is “all facilities”.
- **General service readiness tables** all contain indices: the mean percentage of all items available in facilities, and the percentage of facilities with all items. The denominator is “all facilities”.
- **Service-specific readiness tables** all contain three types of indices: domain indices, the mean percentage of all items available in facilities, and the percentage of facilities with all items. The denominator is “facilities offering the specific service”.
- **Auxiliary indicator tables** do not contain indices, with a few exceptions. The denominator may be “all facilities” or “facilities offering the specific service”.
- **“Special” tables** contain indices based on more complex calculations than the other HHFA indices. These tables are mainly used in the general services readiness dimension. The HHFA contains only a few such tables.
- **Other tables**, mainly used in the management and finance dimension, do not contain any indices and the denominator is almost always all facilities.

The types of HHFA indicator tables, indices, and denominators are summarized in Table 4.

Table 4. Types of HHFA indicator tables

Type of table	Indices	Denominator
1. General service availability	No indices	Mostly all facilities
2. General service readiness	All have indices: mean % of all items, % facilities with all items	All facilities
3. Service-specific availability	No indices (except a few maternal and newborn health services tables)	All facilities
4. Service-specific readiness	All have indices: domain indices; mean % of all items, % facilities with all items	Facilities offering the specific service
5. Auxiliary indicators	Most have no indices; a few have indices similar to readiness tables	All facilities or facilities offering the specific service
6. “Special table”	Have indices with complex calculations; used in general availability and general readiness; indices may be calculated across several tables	
7. Other tables	No indices	All facilities

HHFA indicator inventory platform

The HHFA indicators can be viewed in an online indicator inventory platform (<https://indicator-inventory.hhfa.online/>). An “instructions” tab in the platform provides explanations and short videos on how to use the platform.

The indicator inventory platform provides a quick way of gaining an overview of the HHFA content and the way indicators are organized, as well as providing detailed information on each indicator. Countries planning an HHFA should firstly review the indicator inventory to identify the indicators they need, and then select the relevant questionnaires and questions.

The complete indicator inventory can be downloaded from the platform as an Excel document. Users can also download a selection of indicators or a selection of data fields. An indicator tabulation plan can be generated from the indicator platform in pdf format, showing how the indicators are tabulated and organized in the report outline produced by the HHFA data analysis platform.

The platform contains all the HHFA indicators. Filter options are available to select the indicators related to specific modules and questionnaires. A search function enables the user to find a specific indicator, or a group of indicators related to a specific topic.

The platform provides two views of the indicators: a “listing” view and a “tabulation” view. The “listing” view displays a simple list of the indicators. The “tabulation” view shows a “nested” list that groups the indicators according to the following hierarchy:

Service dimension → Service area → Service subarea → Table → Indicators

There are five service dimensions that broadly correspond to the four HHFA modules:

- Dimension 1. General service availability
- Dimension 2. General service readiness
- Dimension 3. Service-specific availability and readiness
- Dimension 4. Management and finance support systems
- Dimension 5. Clinical quality of care

This system of indicator organization is also used in the HHFA data analysis platform and provides the structure for the HHFA report outline.

Fig. 7 shows a screenshot of part of the indicator platform, illustrating the indicator organization. Clicking on a dimension will display the areas within the dimension. Clicking on an area will display sub-areas, clicking on a subarea will display tables, and clicking on a table will display all the indicators in the table.

Fig. 7. Indicator organization in HHFA indicator inventory platform

Dimension 1. General service availability

Dimension 2. General service readiness

Dimension 3. Service-specific availability and readiness

Area 3.1. Reproductive, maternal, newborn, child, and adolescent health

Area 3.2. Communicable diseases

Area 3.3. Noncommunicable diseases

Sub-Area 3.3.1. Cardiovascular disease

Table 3.3.1.1. Cardiovascular disease service availability

Indicator 3.3.1.1.1. Percentage of facilities offering any services for cardiovascular disease

Indicator 3.3.1.1.2. Percentage of facilities offering diagnosis and treatment of hypertension

Clicking on an individual indicator will display information that describes the indicator, as shown in Fig. 8. This includes the indicator's permanent, unique three-letter ID code (described in Section 3.3) as well as information on the survey questions and code needed to calculate the indicator.

Fig. 8. Example of indicator metadata as displayed in the HHFA indicator inventory platform

Percentage of facilities offering CVD services with blood pressure apparatus ×

Permanent ID	BUN	
Dimension	3.	Service-specific availability and readiness
Area	3.3.	Noncommunicable diseases
Sub-Area	3.3.1.	Cardiovascular disease
Table	3.3.1.2.	Cardiovascular disease service readiness
Indicator	3.3.1.2.3.	Percentage of facilities offering CVD services with blood pressure apparatus

Modules

→ Readiness: Core

Domain

Equipment

Attributes

✓ SARA indicator
✓ PHC indicator

Numerator calculation code

Q6522A_2==1 AND Q6522B_2==1

Q6522A_2 Eqmt in OPD, available: BP apparatus
Q6522B_2 Eqmt in OPD, functioning: BP apparatus

Denominator

Among facilities offering a service

Denominator calculation code

Q7100==1

Q7100 Do providers diagnose and/or manage cardiovascular diseases (CVDs)

Detailed indicator definition

Digital BP machine or manual sphygmomanometer with stethoscope

Notes on data collection

Observed and functional in service site

Dimension 1. General service availability

This dimension describes the presence and implementation of facility-level systems and processes that support the continuous availability and quality of facility services. Indices are not calculated for this dimension.

Service areas

- Health infrastructure
- Health workforce
- Services available

- **Health infrastructure:** contains indicators of facility and bed density, and supplementary information on building structural conditions and accessibility for mobility-limited persons.
- **Health workforce:** contains health workforce density and median indicators for key occupational categories.
- **Services available:** contains information on general outpatient and inpatient services, specific services, and selected diagnostic and treatment procedures offered. (Most of these indicators are also found in Dimension 3. If a country implements the availability module but not the readiness module, the HHFA report produced by the data analysis platform will contain the services available indicators as part of the general service availability dimension of the report. However, if both availability and readiness modules are implemented, the services available indicators are integrated into the service-specific availability and readiness section of the report.)

Note: The infrastructure and workforce density indicators require population estimates as denominators. Therefore, **these indicators can only be calculated when the survey has involved a census of all facilities (public and private) in the country**. Reliable population estimates are also required, usually obtained by projecting census data.

Dimension 2. General service readiness

General service readiness refers to the overall capacity of facilities to provide basic, general health services, based on the presence and functionality of conditions needed for providing the services. An index is generated for each of the five service areas, based on the number of conditions present. An overall general readiness index is calculated based on the mean of the five service areas.

Service areas

- Basic amenities
- Basic equipment
- Standard precautions for infection prevention
- Basic diagnostic capacity (laboratory)
- Essential medicines
- General service readiness index

- **Basic amenities:** assesses power, water source, privacy for consultations, sanitation, communications system, computer with internet and emergency transportation system.
- **Basic equipment:** assesses adult scale, child scale, infant scale, measuring tape, height board, thermometer, stethoscope, blood pressure apparatus, examination light, otoscope, ophthalmoscope and pulse oximeter.
- **Standard precautions for infection prevention:** assesses key items needed for infection prevention and control. Refer to Annex 1 for a detailed list of items. An auxiliary indicator table addresses personal protective equipment.

- **Basic diagnostic capacity (laboratory):** assesses presence of the following onsite testing: haemoglobin; blood glucose; urine dipstick for glucose, protein and ketones; urine test for pregnancy; malaria diagnostic testing; HIV diagnostic testing; and syphilis rapid diagnostic testing.
- **Essential medicines:** assesses a set of general essential medicines that corresponds to the basket of essential medicines for PHC related to SDG 3.8.1. Refer to Annex 1 for a detailed list of medicines.
- **General service readiness index:** each of the five service areas contains a table of indicators representing the presence of basic items. A mean is calculated for each table to provide the readiness index for the service area. This readiness index is the mean percentage of items available among all facilities. A “with all items” index is also provided for each table, indicating the percentage of facilities that have available all the items in the table. In addition, an overall general service readiness index is calculated as the unweighted mean of the five service area readiness indices. Calculation of the readiness indices is described in Annex 1.

Dimension 3. Service-specific availability and readiness

This dimension describes whether health facilities offer a specific service and their capacities to provide that service based on the presence and functionality of tracer items in four domains: trained staff and guidelines, equipment, diagnostic capacity and medicines and commodities.	Service areas ■ Reproductive, maternal, newborn, child, and adolescent health ■ Communicable diseases ■ Noncommunicable diseases ■ Surgical services ■ Emergency services ■ Palliative and rehabilitative care
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Each service area contains one or more **service subareas** that represent a programme or a programme component. Each service subarea contains one or more of the following types of indicator **tables**:

- **Service-specific availability table:** shows the availability of the specific service or service component among all facilities surveyed. The denominator is all facilities.
- **Service-specific readiness table:** contains indicators for a set of items considered essential for the provision of quality care for the specific service. The denominator is the number of facilities offering the specific service. The items are grouped into the four domains: staff and guidelines; equipment; diagnostics; and medicines and commodities. A mean is calculated for each of the four domains and the mean of all the indicators in the readiness table provides the overall readiness index (or score) for the service. A “with all items” index is also produced for readiness tables, indicating the percentage of facilities that have available all the items in the table. Annex 1 describes the calculation of the indices.
- **Auxiliary indicator tables:** there may be one or more auxiliary tables containing further indicators relevant to the programme. The denominator is the number of facilities offering the specific service. Indices are usually not calculated for auxiliary indicator tables.

Fig. 9 shows the availability, readiness and auxiliary indicator tables for family planning services.

Fig. 9. Tables for family planning service availability, readiness and auxiliary indicators**Dimension 1. General service availability****Dimension 2. General service readiness****Dimension 3. Service-specific availability and readiness****Area 3.1. Reproductive, maternal, newborn, child, and adolescent health****Sub-Area 3.1.1. Family planning***Table 3.1.1.1. Family planning service availability**Table 3.1.1.2. Family planning service readiness**Table 3.1.1.3. Family planning auxiliary indicators*

Fig. 10 shows the indicators in the malaria readiness table, including the index for each domain, the overall readiness index for the service and the “with all items” index.

Fig. 10. Malaria service readiness table*Table 3.2.1.2. Malaria service readiness***Indicator 3.2.1.2.1.** Percentage of facilities offering malaria services with guidelines for diagnosis and treatment of malaria**Indicator 3.2.1.2.2.** Percentage of facilities offering malaria services with guidelines for IPTp**Indicator 3.2.1.2.3.** Percentage of facilities offering malaria services with staff trained in malaria diagnosis and treatment**Indicator 3.2.1.2.4.** Percentage of facilities offering malaria services with staff trained in IPTp**Indicator 3.2.1.2.5.** Percentage of facilities offering malaria services with malaria diagnostic testing capacity**Indicator 3.2.1.2.6.** Percentage of facilities offering malaria services with first-line antimalarials**Indicator 3.2.1.2.7.** Percentage of facilities offering malaria services with paracetamol tab/cap**Indicator 3.2.1.2.8.** Percentage of facilities offering malaria services with sulfadoxine-pyrimethamine (SP) tab/cap**Indicator 3.2.1.2.9.** Percentage of facilities offering malaria services with ITN or vouchers for ITN**Indicator 3.2.1.2.10.** Malaria service readiness: mean percentage of staff and guidelines items at facilities**Indicator 3.2.1.2.11.** Malaria service readiness: mean percentage of diagnostics items at facilities**Indicator 3.2.1.2.12.** Malaria service readiness: mean percentage of medicines and commodities items at facilities**Indicator 3.2.1.2.13.** Malaria service readiness: mean percentage of all items at facilities ← **Index: Service readiness index****Indicator 3.2.1.2.14.** Malaria service readiness: percentage of facilities with all items ← **Index: “with all items”****Domain indices**

Dimension 4. Management and finance support systems

This dimension describes the presence and implementation of facility-level systems and processes that support the continuous availability and quality of facility services. Indices are not calculated for this dimension.

Service areas

- Facility governance and management
- Facility finances and accounting
- Systems to support staff
- Systems for staff and patient safety
- Quality monitoring systems
- Health information systems

- **Facility governance and management:** includes facility management committees, support services for routine facility functioning (e.g. staff transport, laundry) and maintenance systems.
- **Facility finances and accounting:** includes budgets, user fees (if relevant) and accountability systems.
- **Systems to support staff:** includes staff credentials, supervision, training, well-being, and benefits.
- **Systems for staff and patient safety:** includes systems for infection prevention and control, cleaning and emergency preparedness.
- **Quality monitoring systems:** includes quality assurances systems and quality monitoring systems for inpatient care, prescribing, adverse reactions to medicines, immunization services, infection prevention and control, and laboratory services.
- **Health information systems:** includes information management and reporting systems, unique identifiers and patient record systems.

Dimension 5. Clinical quality of care

This dimension describes the provider's adherence to standards in the patient care process, as documented in individual patient records. Clinical quality of care indicators are available for each of six¹ service areas: antenatal care, HIV testing and counselling, antiretroviral therapy, preventing mother-to-child transmission of HIV (PMTCT), malaria and tuberculosis.

Service areas

- Antenatal care
- HIV testing and counselling
- Antiretroviral therapy
- Preventing mother-to-child transmission (PMTCT) of HIV
- Malaria
- Tuberculosis

The clinical quality of care dimension is different from the other dimensions in that the data are not collected through the HHFA facility audit methodology, but through a separate record review methodology, using a sample of individual patient/client records per service area (refer to Section 4.3 for further details).

¹ Record review questionnaires and indicators may be developed for additional service areas in the future.

For each service area, there are indicators representing various steps in the care process that can be used to inform quality improvement strategies. An index is also calculated for each individual patient to represent the complete care process (comprised of a package of assessment, diagnosis and treatment). This enables assessment of the quality of a complete service, as well as the individual components of the care process. Finally, a facility average score or index is calculated for each service, based on the mean of the individual patient indices.

The indicators were developed in collaboration with subject-matter experts and aligned with standards of care for each service area. Countries may need to adapt the indicators to ensure that they align with the service packages used in the country. However, efforts should be made to maintain the standard HHFA indicator definitions and calculations, to facilitate interpretability of results as well as comparisons over time and across countries. If necessary, country-specific additional indicators can be added, rather than substantially modifying the standard HHFA indicators.

The sampling method for the HHFA record review was developed to provide a picture of the routine care process for a specific service at the facility level. Analysis is carried out using the facility as the unit of measure¹ and is based on the facility-level index, and not on the individual patient indices. When using the facility as the unit of measure, facility averages for results are calculated, and subnational or national results represent the averages across facilities. The results are weighted so that the facility results are representative of the facility distribution in the country. (This weighting method is also used for the facility audit part of the HHFA.) This ensures that the practices in facilities with smaller caseloads (often remote and less-supervised facilities) are proportionally represented – an important consideration when working to ensure equitable access to quality services. Using the facility as the unit of measure may also provide information that helps to identify management and supervision issues that are present across the service.

If a record review has been conducted in conjunction with an HHFA facility audit that measures service availability and readiness, efforts should be made to combine the findings from both assessments to provide an in-depth, meaningful understanding of the service. If the record review reveals that certain processes have not been conducted (as evidenced by lack of recording in the patient record), it is important to know whether the items required to conduct the processes were available or not. For example, failure to conduct a malaria blood test may be the result of a provider not following protocols, but also may occur because the test is not available in the facility. Combining service availability, service readiness, and technical care process data provides a unique opportunity to deepen understanding of the challenges and potential intervention points for improving health service quality.

3.6 HHFA data analysis platform

The HHFA data analysis platform is an online web application that automatically calculates the HHFA indicators, and displays them in tables and graphs, as well as providing the outline of the HHFA report. The analysis platform can be used to undertake all the analysis required to produce the HHFA report for each indicator. A country team creates a country-specific version (or “instance”) of the analysis platform and decides who will be given access to it. The team can then work collaboratively in the platform on the analysis of the HHFA dataset and report.

¹ The HHFA uses the facility as the unit of measure (rather than the individual patient/client) for the following reasons: 1) individual-level analysis provides information on the quality of care received by the majority of patients/clients, but may mask issues in low-volume facilities which are often isolated or under-resourced; 2) individual-level analysis requires data on facility caseloads which would require a level of time and effort not feasible in the context of the HHFA; and 3) the HHFA aims to evaluate the care process rather than an individual service provider, as the care process is frequently influenced by factors other than provider expertise, such as availability of specific tools or commodities to provide a service and by service expectations of supervisors at facility or higher level. However, as a future development, WHO may also explore options for using the individual as the unit of analysis for an in-depth assessment.

The platform can be accessed at <https://analysis-platform.hhfa.online>. Anyone with a WHO e-mail address (@who.int), or who has already been given guest access to the WHO sign-in system, can access the platform by navigating to the website and clicking “Sign in”. Country team members who do not have WHO guest access should contact the HHFA coordination team at hhfa@who.int to request access. An invitation will then be sent to the team members which will grant them access to the platform through the WHO single sign-in system (Microsoft SSO). Once an invitation has been accepted and access has been granted, users can navigate to the website and click “Sign in” to start using the analysis platform.

The data analysis platform is designed to make it as easy and quick as possible to analyse an HHFA dataset. Ideally, a team uploads a cleaned dataset (cleaning completed in CSPro), configures a minimal set of options in the platform, runs all the analyses required to produce the default (standard) indicators and tables, and generates a template report in Word (and PDF) within 1–2 hours. A dataset that has been collected using the official standard HHFA CSPro package, and has been vetted and cleaned, should not require significant configuration on the platform. A dataset collected using a customized tool, or with renamed questions or variables, will require some additional configuration and additional time. Also, any country-specific indicators will need to be added to the platform.

The platform runs fully online and does not require any software to be downloaded or installed. The user can simply navigate to the website and start to use the tool.

Also, the platform does not require users to be familiar with any statistical software (R, Stata, SPSS, SAS, etc.). However, users will need a foundational understanding of descriptive statistics to be able to interpret and interrogate the platform’s analytical outputs, and to configure any required custom indicators.

How the HHFA data analysis platform works

The platform works by running analysis scripts that are auto-generated from indicator-specific codes in the HHFA indicator inventory, and from information that the platform infers from the uploaded dataset. Using the pre-established code for each indicator’s numerator and denominator, and the list of questions and variables in the dataset, the platform internally creates an analysis script to produce individual tables and figures. These scripts are then run on the dataset, and the outputs are presented to the user in the form of tables and graphs. The basic process for using HHFA analysis platform is described in Box 2.

Box 2. How to use the analysis platform

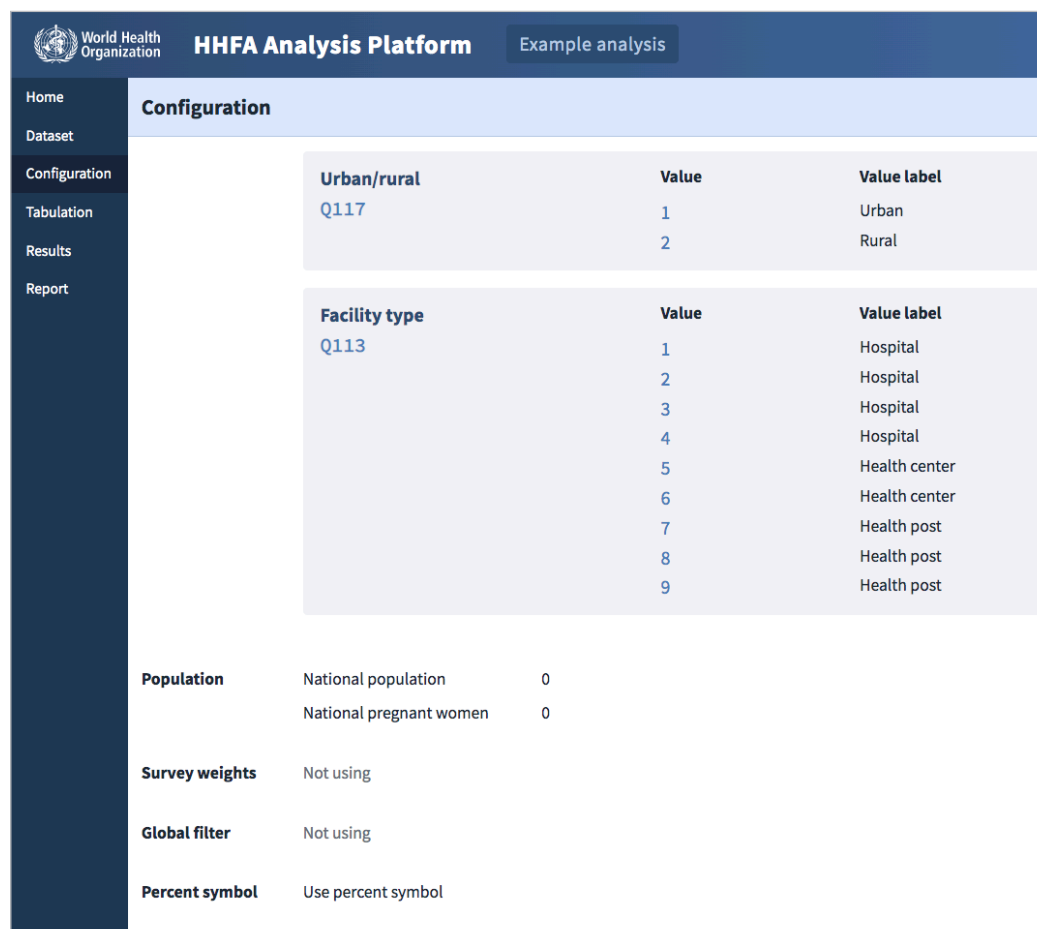
1. Login to the platform and create a “version” of the report that the team aims to prepare.
2. Upload a cleaned dataset.
3. View the dataset (e.g. Fig. 11), checking that all variables exist as expected, and remap any variables that have different names to those used in the HHFA indicator inventory.
4. Choose the variables that you want to use as “stratifiers”, e.g. for administrative region, managing authority, urban/rural, facility type (e.g. Fig. 12).
5. If the dataset represents a census, enter information on population numbers, to enable the platform to calculate density indicators.
6. Import the set of indicators (from the HHFA indicator inventory) that you want to calculate for your analysis and report, and that are appropriate for your dataset (depending on the modules and core and additional questions that were used for data collection).
7. Add, remove, or customize indicators as needed (e.g. Fig. 13).
8. “Run all” tables (which may take some time to complete), or individually click “Run” for each table. Check that each table has run successfully and has been produced. You will be able to see the table on the screen and produce related charts and figures (e.g. Figs 14 and 15).
9. Once all tables have been produced, you have the option to generate a Word document, with all tables contained in it, or to print a PDF version of all tables.

For users who may want additional control over the statistical analyses, there are options to view and check the indicator code and the underlying scripts, and to download the intermediate outputs that are created as part of the analyses (for example, indicators calculated at facility-level, before aggregation).

All the codes for indicator calculation can be seen and edited within the platform. Users are encouraged to interrogate these codes and ensure they align with their dataset. Users are also able to download the auto-generated analysis scripts and run them offline, outside of the platform, using R or Stata.

Fig. 11. Viewing the uploaded dataset

 HHFA Analysis Platform Example analysis HHFA Tools						
Home	Dataset			EXAMPLE_DATASET_HHFA.dta (11/11/2022, 03:13:12)		
Dataset					Clear dataset	Remap variables
Configuration	Question number	Label	Unique responses		Missing	Type
Tabulation	Q100	Facility Code	3179	[]	0	numeric
Results	Q101	Supervisor validation	1	[2]	0	double
Report	ID_TEAML	Team Leader	68	[]	0	numeric
	WGT	Facility Weight	1	[]	All missing	numeric
	Q102	Facility Name	3128	[]	0	character
	Q103	Facility known by other name	2	[1,2]	0	double
	Q103T	Q103 if other, specify	120	[]	0	character
	Q104	Facility location	2	[1,2]	0	double
	Q105	Region/Province code	10	[11,12,13,14,15,16,17,18,19,20]	0	numeric
	Q105T	Region/Province name	10	["Central","Copperbelt","Eastern","Luapula","Lus...	0	character
	Q106	District code	17	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,80]	0	numeric
	Q106T	District name	117	[]	0	character
	FVISIT	Final Visit	1740	[]	0	character

Fig. 12. Configuring the stratifier variables


HHFA Analysis Platform Example analysis

Configuration

Urban/rural	Value	Value label
Q117	1	Urban
	2	Rural

Facility type	Value	Value label
Q113	1	Hospital
	2	Hospital
	3	Hospital
	4	Hospital
	5	Health center
	6	Health center
	7	Health post
	8	Health post
	9	Health post

Population

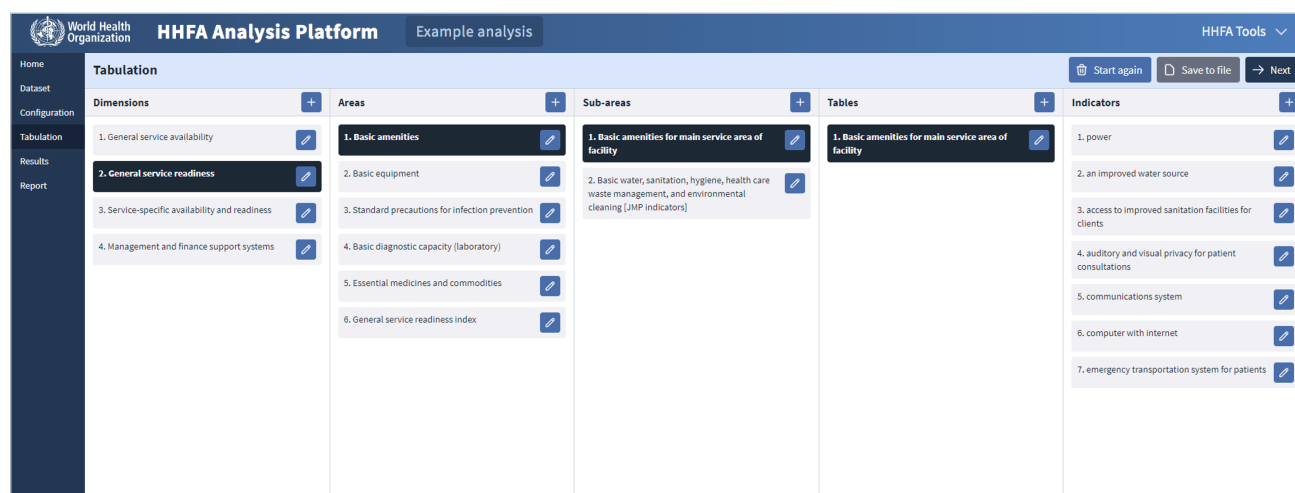
National population 0

National pregnant women 0

Survey weights Not using

Global filter Not using

Percent symbol Use percent symbol

Fig. 13. Customizing indicators


HHFA Analysis Platform Example analysis HHFA Tools

Tabulation

Start again Save to file Next

Dimensions	Areas	Sub-areas	Tables	Indicators
1. General service availability	1. Basic amenities	1. Basic amenities for main service area of facility	1. Basic amenities for main service area of facility	1. power
2. General service readiness	2. Basic equipment	2. Basic water, sanitation, hygiene, health care waste management, and environmental cleaning [JMP indicators]		2. an improved water source
3. Service-specific availability and readiness	3. Standard precautions for infection prevention			3. access to improved sanitation facilities for clients
4. Management and finance support systems	4. Basic diagnostic capacity (laboratory)			4. auditory and visual privacy for patient consultations
	5. Essential medicines and commodities			5. communications system
	6. General service readiness index			6. computer with internet
				7. emergency transportation system for patients

Fig. 14. “Running” individual table analyses to see outputs

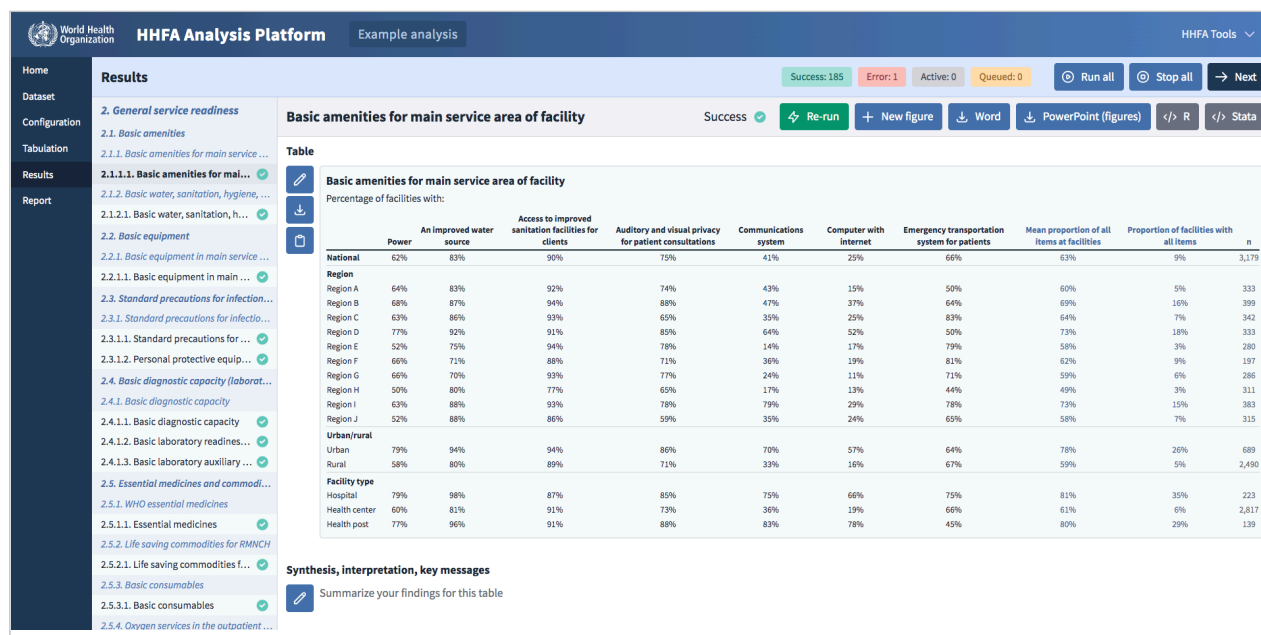


Fig. 15. Options for customizable charts

